

View Reviews

Paper ID

259

Paper Title

Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations

Track Name

WSDM2025 Main

Reviewer #1

Questions**1. Summary and contributions: Briefly summarize the paper and its contributions.**

The paper studies a cold start user-item recommendation setup. The main idea is to use keywords for candidate generation, followed by re-ranking of the candidates by an LLM (using the keywords as context). The authors propose an unsupervised method (message passing on graph - MPG) based on user-keyword and keyword-item relationships to assist the candidate generation process. Through offline evaluations using off the shelf LLMs (Gemini, GPT 3.5, Mistral and Llama) the authors show the effectiveness of their approach.

2. Strong points. List three to five strong points about the paper. Please be precise and explicit. Clearly explain the value and nature of the contribution.

1. The paper is well written and easy to follow.
2. The MPG method is intuitive, and contributes to lifts in precision and recall in offline evaluation.
3. The authors have shared their prompt engineering effort (e.g., zero shot, few shot strategies in their setup) and learnings from it could be useful for broader LLM based recommendation setups.
4. The results cover 4 off-the-shelf LLMs.

3. Weak points. List three to five weak points about the paper. Please clearly indicate whether the paper has any mistakes, missing related work, or results that cannot be considered as a contribution. Please be polite, specific, and constructive.

1. The modeling contributions in the paper are weak. MPG seems like the only new contribution from a modeling standpoint, and although it provides lifts in precision and recall vs CBR, more analysis/ablation study is needed to understand the reason for the lifts (e.g., what is the efficacy of TF-IRF?).
2. Apart from the candidate generation part, the remainder of the paper is mostly an empirical study with off-the-shelf LLMs. Claude is missing from the LLMs list and its

context size of 100K+ can affect the conclusions in the study (e.g., providing all reviews vs extracting keywords).

3. The baseline comparison in RQ2 needs more depth, i.e., more baseline approaches are needed to compare against the keyword approach. For example, if a summary of the reviews was used instead of all reviews, would the task become easier for LLMs?

4. Online results are not present. The scalability and cost considerations for using the proposed method for real time recommendations is not discussed in the paper.

4. Detailed Evaluation. Please provide detailed feedback about the strengths and the weaknesses of the paper and support your overall rating. You may talk about significance, technical depth, novelty, reproducibility, relevance to the community, and potential ethical considerations. Please be polite, specific, and constructive.

The paper is mostly an empirical study with off the shelf LLMs. Although the results are insightful, limited modeling contributions and missing baselines (e.g., Claude LLM, using review summaries as discussed in the weakness section) leave a gap in the paper's assessment. Also online results are missing which could have shed light on the approach's feasibility for real time recommendations.

5. Overall rating.

Weak Reject (Probable reject, unless convinced otherwise)

8. Reviewer confidence.

Knowledgeable: I have read papers on this topic.

Reviewer #2

Questions

1. Summary and contributions: Briefly summarize the paper and its contributions.

This paper proposed a framework to address the cold-start problem in recommendation systems by utilizing keyword-driven retrieval and LLMs for re-ranking. The framework is rigorously evaluated, demonstrating improved performance compared to traditional baselines.

2. Strong points. List three to five strong points about the paper. Please be precise and explicit. Clearly explain the value and nature of the contribution.

1. The paper proposed the keyword driven retrieval and LLM-based re-ranking framework, which provides a novel and effective solution to cold-start recommendation.

2. The extensive experiments demonstrate the effectiveness of the proposed framework.

3. The integration of LLMs in re-ranking stage adds robustness to the framework, by capturing complex user preference from minimal input.

3. Weak points. List three to five weak points about the paper. Please clearly indicate whether the paper has any mistakes, missing related work, or results

that cannot be considered as a contribution. Please be polite, specific, and constructive.

1. The paper briefly mentions the issue of hallucinations in LLMs, but does not explore this in sufficient depth. LLMs are prone to such issues, especially in recommendation setting. Add more discussion about the issues and addressing potential risks would be beneficial.

2. The baselines, especially the retrieval baselines are fairly traditional. Including comparisons with more advanced methods, e.g., use the embeddings from LLMs, to do the retrieval would provide more insights of the framework's performance.

3. The keywords is central to the proposed framework, which is fully relying on the tool like SpaCY. More discussions about the keyword quality or alternative extraction methods would strengthen the paper.

2.

4. Detailed Evaluation. Please provide detailed feedback about the strengths and the weaknesses of the paper and support your overall rating. You may talk about significance, technical depth, novelty, reproducibility, relevance to the community, and potential ethical considerations. Please be polite, specific, and constructive.

Please see the strong points and weak points.

5. Overall rating.

Weak Accept (Probable accept, unless convinced otherwise)

8. Reviewer confidence.

Expert: I have published papers on this topic.

Reviewer #3

Questions

1. Summary and contributions: Briefly summarize the paper and its contributions.

This paper considers the use of keyword information to enhance the performance of recommendation systems for cold users and designs a keyword-based prompt to utilize Large Language Models (LLMs) for ranking candidate items. The contribution of this paper lies in the design of an algorithm that leverages keyword information and LLMs to address the cold start problem, providing a novel approach to tackle this issue.

2. Strong points. List three to five strong points about the paper. Please be precise and explicit. Clearly explain the value and nature of the contribution.

1. The idea of using keywords to mitigate the cold start problem is interesting, and this method also aligns well with the needs of real-world scenarios.

2. Comprehensive Analysis Experiment: This paper designs an analytical experiment to effectively support the motivation for using the keyword instead of reviews,

demonstrating the superiority of using the keyword compared to using the review. Additionally, for the re-rank module, this paper presents experimental results from different LLMs, indicating the good generalizability of using LLMs for re-ranking.

3. Clear Expression: The writing in this paper is easy to understand, and the illustrative diagrams are quite intuitive.

3. Weak points. List three to five weak points about the paper. Please clearly indicate whether the paper has any mistakes, missing related work, or results that cannot be considered as a contribution. Please be polite, specific, and constructive.

1. Confusion in Mathematical Notation: There is considerable confusion regarding the mathematical symbols used throughout the paper. for example, the term $\hat{y}_{u,r,p,r_n}$ in Equation (3) is not formally defined; the summation over all items $r \in \{\mathcal{R}\}$ in Equation (5) might be incorrect; in the line above Equation (6), a^{wr} should probably be a^w_r ; the definitions and distinctions between a_r^w and $\mathcal{A}_r^w$ in the text below Equation (5) and later in the paper are ambiguous; the definition of w in Equation (7) is unclear, as w represents a term and r represents a node in the text, yet in $\mathcal{A}_r^w \in \mathbb{R}^{w \times r}$ and $\mathcal{M}_w \in \mathbb{R}^{1 \times w}$ below Equation (7), both are represented as real numbers. The vector cross product in Equation (7) should likely be a dot product.

2. Unclear Motivation. There is a need for a more detailed explanation of the importance of structured information that introduced by MPG module. Besides, the application of the attention mechanism in the CBR module have been widely used in previous research [1, 2], but there is a lack of explanation for the motivation behind using these technologies in the context of this paper's problem.

3. Lack of detailed explanation regarding the relationship between the modules. The relationship between the MPG and CBR modules is ambiguous, leaving readers uncertain whether these modules are intended to be used independently or in conjunction.

4. Lack of Comparative Experiments and Limited Dataset: The paper does not include comparative experiments with other methods addressing cold start users, such as those proposed by [3, 4, 5]. Moreover, the study is conducted on only two scenario datasets, which is insufficient to demonstrate the generalization and universality of the proposed method. Including a broader range of datasets and comparative analyses would strengthen the paper's claims.

[1] Xu, Wentao, et al. "Enhancing Content-based Recommendation via Large Language Model." arXiv preprint arXiv:2404.00236 (2024).

[2] Zuo, Chandler, et al. "Inductive Modeling for Realtime Cold Start Recommendations." Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. 2024.

[3] Wei, Yinwei, et al. "Contrastive learning for cold-start recommendation." Proceedings of the 29th ACM International Conference on Multimedia. 2021.

[4] Lin, Xixun, et al. "Task-adaptive neural process for user cold-start recommendation." Proceedings of the Web Conference 2021. 2021.

[5] Lee, Hyeop, et al. "Melu: Meta-learned user preference estimator for cold-start recommendation." Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. 2019.

4. Detailed Evaluation. Please provide detailed feedback about the strengths and the weaknesses of the paper and support your overall rating. You may talk about significance, technical depth, novelty, reproducibility, relevance to the community, and potential ethical considerations. Please be polite, specific, and constructive.

In conclusion, while this paper studies on an interesting problem, this work has some limitations on mathematical notation, unclear motivation for certain modules, and a lack of comparative experiments and dataset diversity.

Addressing these weaknesses would greatly enhance the paper's impact and reliability.

Overall, the paper makes a valuable contribution to the field, but requires further refinement to fully realize its potential.

5. Overall rating.

Weak Reject (Probable reject, unless convinced otherwise)

8. Reviewer confidence.

Knowledgeable: I have read papers on this topic.

Reviewer #4

Questions

1. Summary and contributions: Briefly summarize the paper and its contributions.

The authors claim two points: 1) that LLM-based recommendation can potentially alleviate the issue of cold-start users, yet run into the hallucination problem without fine-tuning; and 2) that previous user-user content-based methods that tackle the cold-start user issue raise privacy concerns. To this end, the authors propose a preference keyword-based recommendation framework (KALM4REC) as a practical approach for handling the issue of cold-start users. KALM4REC consists of 2 steps: 1) retrieving relevant item candidates based on keywords provided by users, and 2)

leveraging LLMs to rerank the candidates from the previous step. KALM4REC is a novel framework that puts an emphasis on keywords when obtaining candidates for recommendation, and outperforms baselines across 4 real-world datasets.

2. Strong points. List three to five strong points about the paper. Please be precise and explicit. Clearly explain the value and nature of the contribution.

1. The authors suggest a novel approach that integrates keyword-based retrieval and LLMs for recommendation.

2. Various real-world datasets are used for evaluation, demonstrating the applicability of the proposed framework.

3. Extensive experiments provide a thorough exploration of the performance of KALM4REC.

3. Weak points. List three to five weak points about the paper. Please clearly indicate whether the paper has any mistakes, missing related work, or results that cannot be considered as a contribution. Please be polite, specific, and constructive.

1. Major claims used as motivation for the paper are presented without sufficient explanation or citations, and some are not further explored in the experiment section.

2. Poor writing; there are many ambiguous terminologies and notations, and confusing/missing explanations in the methodology and experiment section make it hard to understand key components.

3. Lack of appropriate baselines.

4. Detailed Evaluation. Please provide detailed feedback about the strengths and the weaknesses of the paper and support your overall rating. You may talk about significance, technical depth, novelty, reproducibility, relevance to the community, and potential ethical considerations. Please be polite, specific, and constructive.

While the authors claim that utilizing a few input keywords from users is a practical way to alleviate the cold-start user issue, there are no references or real-world examples that illustrate this claim.

The hallucination problem is presented as a major motivation for the proposed method, yet this issue is not mentioned after the introduction and related works section. An experiment demonstrating that KALM4REC tackles this issue should be included in order to strengthen this motivation.

The claim that the CBR module fails to encode collaborative signals requires further explanation. On that note, CBR is shown to be an inferior retrieval module compared to MPG; this brings into question why CBR is used in this paper at all. Either only including MPG or further illustrating the connection between CBR and MPG would be

beneficial to the overall flow of the paper.

In MPG, the user-keyword interactions are not used during message passing and are only utilized in the form of a matrix when obtaining item scores. As such, it seems like a stretch to use the term "tripartite graph structure" when the structural information of user-keyword interactions are not needed.

How the keywords for test users are obtained needs to be more clearly illustrated. If they are obtained from reviews by test users, the number of such reviews should be provided, i.e., if "all" reviews are utilized, this would not be aligned with the problem setting of cold-start users.

The baselines used for comparison (MF, MVAE) need further explanation as to how they are implemented. How are they adapted to handle the cold-start user problem? On this note, baselines that are explicitly designed for the cold-start user issue, e.g., popularity or content-based methods, should be included for comparison.

Examples of poor writing and missing information:

- "Despite these advances, there remains a significant gap" in the related works section: what gap are the authors referring to? If this simply means that previous methods do not utilize keywords, there should be more evidence or motivational examples that illustrate why it is a problem that needs to be addressed.
- In the CBR section, T is used both as the number of keywords and as a matrix transpose notation. Additionally, there should also be equations related to the construction of item embeddings. Further, while all items are restaurants in the datasets used for experiments, the term "item" should be consistently used in the methodology section.

5. Overall rating.

Reject (Will argue strongly to reject)

8. Reviewer confidence.

Expert: I have published papers on this topic.

Reviewer #5

Questions

1. Summary and contributions: Briefly summarize the paper and its contributions.

This paper proposes a framework for cold-start recommendation with keywords and large language models. The core of the framework lies in the representation learning of extracted keywords from reviews. The framework uses the user's keywords to retrieve candidate items and leverages LLMs to re-rank the candidate items for final recommendation. The experimental evaluation includes two datasets and shows consistent improvements by the proposed framework.

2. Strong points. List three to five strong points about the paper. Please be precise and explicit. Clearly explain the value and nature of the contribution.

1. The proposed framework is novel. This is the first work to explore keywords with LLMs for cold-start recommendation.
2. The paper is well-organized and easy to follow. The figures in the paper is illustrative.
3. The experiments indicate the effectiveness of the proposed framework on two commonly used datasets. There are comparisons on different foundation models used for re-ranking.

3. Weak points. List three to five weak points about the paper. Please clearly indicate whether the paper has any mistakes, missing related work, or results that cannot be considered as a contribution. Please be polite, specific, and constructive.

1. A previous work also studies the keywords for cold-start recommendation, but is not discussed in the work. (Ji, Ke, and Hong Shen. "Addressing cold-start: Scalable recommendation with tags and keywords." Knowledge-Based Systems 83 (2015): 42-50.)
2. The evaluated model from OpenAI seems outdated compared to other LLMs. E.g.
3. It seems that the keywords for users are extracted from reviews. However, in real-world applications, the keywords are not likely to be as rich as extracted by their review. This could be potentially a blocker for real-world adoption.

4. Detailed Evaluation. Please provide detailed feedback about the strengths and the weaknesses of the paper and support your overall rating. You may talk about significance, technical depth, novelty, reproducibility, relevance to the community, and potential ethical considerations. Please be polite, specific, and constructive.

The paper proposed a novel framework and studies an important problem in recommendation systems. The evaluations show that the framework achieves better performance compared to baselines. The code is open-sourced for reproducibility and additional study on keyword extractors is provided in supplementary materials. Although some weak points need to be addressed, this paper presents a solid contribution to the community.

5. Overall rating.

Weak Accept (Probable accept, unless convinced otherwise)

8. Reviewer confidence.

Knowledgeable: I have read papers on this topic.